

University of Groningen

Evolvability and organismal architecture

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DOI:
[10.33612/diss.864174360](https://doi.org/10.33612/diss.864174360)

IMPORTANT NOTE: You are advised to consult the publisher's version (publisher's PDF) if you wish to cite from it. Please check the document version below.

Document Version
Publisher's PDF, also known as Version of record

Publication date:
2024

[Link to publication in University of Groningen/UMCG research database](#)

Citation for published version (APA):

Tiso, S. (2024). *Evolvability and organismal architecture: The blind watchmaker and the reminiscent architect*. [Thesis fully internal (DIV), University of Groningen]. University of Groningen.
<https://doi.org/10.33612/diss.864174360>

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Propositions

Accompanying the dissertation

Evolvability and Organismal Architecture

the Blind Watchmaker and the Reminiscing Architect

by

Stefano Tiso

1. Dawkins (1986) compared adaptive evolution with a “blind watchmaker”. This traditional view of evolution as a myopic process neglects that knowledge about the environment can be deeply ingrained within organismal architecture making it more similar to a “reminiscent architect”.
2. Organisms that survived the constantly shifting challenges faced in evolutionary history must have evolved structures and features that make them highly evolvable. Yet, evolvability is hardly ever considered in evolutionary analyses.
3. Considering the mechanisms underlying evolvability helps clarify misunderstandings and debate in the literature and opens unexplored directions of research (Chapter 2 of this thesis).
4. Architectural features leading to an overrepresentation of selectively favoured mutations, and hence higher evolvability, readily evolve even in simple models (Chapter 3 of this thesis).

5. Even when only sporadically selected and subject to mutational decay, complex adaptations can evolve and be maintained over long periods. (Chapter 4 of this thesis).
6. Evolutionary biology can enrich the computational sciences; vice versa, insights from evolutionary computation can shed new light on biological concepts like evolvability (Chapters 1 and 5 of this thesis).
7. Single definitions of multifaceted concepts such as evolvability hardly help in pinpointing the nature of the phenomenon, they will help, however, in highlighting what the giver of the definition finds particularly interesting.
8. Each perspective has its limitations. It is therefore detrimental to exclusively argue for the validity of one over the others. Instead one should strive to understand all of them to have a varied toolkit at one's disposal when tackling new problems.
9. Theoretical models should not only serve as digital experiments aimed at mimicking reality but also as computational metaphors that explore new ideas on how nature works.
10. Science is a human endeavour, and as such it is primarily driven by personal taste. Science is not poetry simply because scientists choose to engage with the world rigorously and not evocatively.
11. We know a bit more than 100 years ago and will know a bit more in 100 years. What makes you want to stay is the will to contribute one step forward in this long marathon into the universe.